

To group or not to group? Group sizes for requirements elicitation

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ABSTRACT

Context: Requirement elicitation can be done by individuals or by groups. Computer-based system development life-cycle models suggest having people working together for many steps. Also, recommendations about analysis and design methods indicate that some processes could take advantage of group work. In requirements engineering, groups are suggested for requirements elicitation.

Objectives: From the software and the requirements engineering viewpoints, and in turn for companies, a relevant overall research question is “What is a suitable size for a requirements elicitation group?” Our goal was to answer this question, first by looking for available guidelines in textbooks and secondly by investigating requirements elicitation in companies.

Method: To address the research question, we conducted two studies. The first was a review of most widely adopted software and requirements engineering textbooks. The second was a study aimed at identifying factors affecting group size for requirements elicitation, based on an online questionnaire submitted to professional analysts.

Results: The review of the textbooks showed that very few give advice on the number of analysts to involve in requirements elicitation sessions. When they do, guidelines are quite general and not supported by empirical data. According to data gathered from the questionnaire, most companies use and suggest using small groups. Data also allowed identifying four categories of factors useful to make decisions about requirements elicitation group sizes: people, relation, project, and output.

Conclusion: Both the textbook review and the data from the questionnaire say that it is better to aim for small groups than to have individual analysts working separately. The ideal number of analysts for a requirements elicitation session appears to be 2, but large groups are necessary in some cases. Factors in all the four categories have to be considered in deciding the size of groups.

1. Introduction

Whether a process should be carried out by a group or by an individual, and if by a group, how big is the group, is called the GvsI issue in this paper. The group vs individual (GvsI) issue is relevant in many disciplines, e.g., systems theory, psychology, psychiatrics, etc. [46,53,57]. Managerial disciplines have investigated the GvsI issue in many contexts, e.g., in project management, operations research, decision making, etc. [20,45,51].

In software engineering (SE), some computer-based system (CBS) development life-cycle models suggest having people working together for specific tasks [49]. Examples include Joint Application Design (JAD) [9] and Scrum meetings [47]. Also, recommendations about analysis and design methods suggest that some processes could take advantage of

group work.

In requirements engineering (RE), two key processes in which groups are often used, groups of stakeholders, groups of business or requirements analysts (for the rest of the paper “analysts”), or both, are requirements elicitation and requirements idea generation. *Requirements elicitation* is the general process of eliciting a computer-based system’s requirements from the organization of the CBS’s client. By “organization” is meant not only the client’s people, but also its environment, its culture, its current work processes, and its current CBSs, [18,37]. *Requirements idea generation* is the specific process of having people produce ideas for requirements of a CBS. Requirements idea generation often benefits from creativity making use of brainstorming [36] or other creativity techniques [16,25,40].

This paper collects all RE processes that are considered requirements

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elicitation, requirements idea generation, or both under the term “ReqElic”. For ReqElic, in the literature there are recommendations both to use groups and to use individuals. However, often, no data are given to support any claim for how to decide whether a process should be done by a group or by an individual and if by a group, how big the group should be. In addition, little information is given about the factors for making the decision. See Section 6, about related work for more details.

This gap in the existing literature suggested the research question that is the focus of this paper is, in its most general form:

“What is a suitable size for a requirements elicitation group?”

To address the research question, we conducted two studies. The goal was to combine knowledge from the literature with real-world data from practitioners. For the first study, we propose a scenario in which a requirements analyst is looking for information on the group size for a given project. Based on the observation that practitioners rarely if ever consult the research literature [3], we assumed that he or she will first check his or her old SE and RE textbooks. Accordingly, we identified the most widely adopted SE and RE textbooks and reviewed how they deal with the GvsI issue for ReqElic.

The second study aimed at gathering data on the GvsI issue for ReqElic, identifying factors affecting group size for requirements elicitation in computer-based system development projects, based on an online questionnaire submitted to professional analysts.

Our studies give a preliminary answer to the research question, supporting with empirical data recommendations found in SE and RE textbooks. Both the textbooks review and the data from the questionnaire suggest that it is better to aim for small groups. However, the review confirms that textbooks give, if any, very general advice about the number of analysts to involve in ReqElic. In addition, this advice is usually not supported by empirical data. According to the results of the second study, the ideal number of analysts for a requirements elicitation session appears to be 2, but large groups are necessary in some cases. Factors mentioned as relevant in deciding the size of groups for ReqElic have been classified into four categories: project, people, relations, and output. Besides those characterizing a system project (project), there are factors related to characteristics of the analysts (people), the way they work in group (relations), and characteristics of the elicited requirements (output). Results of the questionnaire highlighted interesting differences for the role of the four categories as the reasons to decide on adopting a group in ReqElic and for the choice of the size of the group.

Some preliminary results of the survey are given in three reports, each using the filled-in questionnaires received by its publication. When we had 35 filled-in questionnaires, we used the data as corroborating data into a workshop paper about controlled experiments for deciding the GvsI issue with respect to one particular creativity enhancement technique, POEPMcreate [43]. By the time we had enhanced this workshop paper for journal publication, we had received 53 filled-in questionnaires. A similar analysis of these data was incorporated as the corroborating data into the journal paper [44]. When we had 92 filled-in questionnaires and had hopes that we might receive more, we analyzed the data we had for a magazine paper about just the first part of the questionnaire [29]. All these analyses had goals that were more limited than those of this paper, which includes also a review addressing the GvsI issue in SE and RE textbooks and an analysis of the open questions, which were not analyzed before.

In the rest of the paper, Section 2 consists of a review of well-known SE and RE textbooks, describing how well they addresses the GvsI issue. Section 3 introduces the design and submission of the on-line questionnaire, and illustrates the results. Section 4 discusses the results of the two studies. Section 5 considers the limitations of and the threats to the validity of the conclusions. Section 6 discusses related work, and Section 7 concludes the paper with guidelines and future work.

2. The group vs. individual problem in the textbooks

The scenario that drives decisions concerning the conduct of the first study to review the most successful SE and RE textbooks is that when an analyst needs advice on staffing an RE process and to deal with the GvsI issue for this process, he or she will consult his or her old textbooks. Another reason to check textbooks is that the successful ones are continuously updated with best practices from the industrial and academic worlds.

The goal of the review of any textbook was to see if it takes into consideration the fact that ReqElic activities can be accomplished as group or individual activities. If the answer to that question is “Yes”, then the goal was to see what recommendation the book made in the GvsI issue.

Section 1 explains that we decided to identify the most widely adopted SE and RE textbooks to see how they deal with the GvsI issue for ReqElic. We decided to use the number of editions and the number languages to which the textbooks have been translated as the measure of their popularity.¹

Available textbooks and related information were retrieved by the first author using the Google (google.com), Amazon (amazon.com), and Worldcat (worldcat.org) search engines. Strings used for the iterative and incremental search process are “software engineering”, “software” for SE textbooks, and “requirements engineering”, “software requirements”, “requirements” for RE textbooks. Filters offered by the three websites to choose the language (English) and the availability of a printed version allowed to reduce the high number of outputs. Amazon recommendations were also useful, as for a given book, information about similar books is visualized. Worldcat allows finding libraries, mostly university libraries, in which a given book is available, and was used to gather data about the date of the first edition, the number of editions, the languages to which the textbook has been translated. Many of the results confirmed the personal knowledge of the authors of this paper about the most popular SE and RE textbooks.

Further criteria, different for SE and RE were then applied for a matter of feasibility of the review to deal with the tradeoff of an in-depth analysis of the content related to the GvsI and the effort required to review many textbooks, each with an average of over 400 pages.

2.1. Software engineering textbooks

2.1.1. Choosing the SE textbooks

For selecting the most popular SE textbooks among those retrieved, we considered those that

- have reached at least three editions, AND
- have been translated into at least one language other than its original one, which is usually English.

We also took into account feasibility issues, such as the number of versions and of pages we needed to examine.

We found five qualifying textbooks, because the sixth candidate had only two editions and two translations. They are listed in Table 1 ordered by decreasing numbers of editions. The information in column 2 is a citation to the most recent edition of the textbook that we know of (Since each table entry is citationally complete, there is no entry for it in the bibliography). More than one citation for a textbook, means that it has undergone a title or an author-list change. Column 3 gives the dates of its first and last editions and the languages to which it has been translated.

¹ For example, Pressman’s *Software Engineering: A Practitioner’s Approach* has been used at over 500 Universities worldwide and is also widely used in industry and is required reading for many industry and government courses (https://en.wikipedia.org/wiki/Roger_S._Pressman).

Table 1
Software engineering textbooks.

Textbook	Editions
1 Sommerville, Ian: <i>Software Engineering</i> , 10th ed., Pearson, 2018	1st ed. 1982 10th ed. 2015 International Edition, 10th ed. 2018 Translated to Chinese, Czech, Dutch, French, German, Greek, Hungarian, Korean, Italian, Japanese, Polish, Portuguese, Spanish, Thai
2 Pressman, Roger S. and Maxim, Bruce R.: <i>Software Engineering: A Practitioner's Approach</i> , 9th ed., McGraw Hill Education, 2019 Pressman, Roger S.: <i>Software Engineering: A Practitioner's Approach</i> , 7th ed., McGraw Hill Education, 2010	1st ed. 1982 9th ed. 2019 International Student Edition, 9th ed. 2020 Translated to Chinese, German, Korean, Italian, Japanese, Polish, Portuguese, Spanish
3 Schach, Stephen R.: <i>Object-Oriented and Classical Software Engineering</i> , 8th ed., McGraw-Hill, 2011 Schach, Stephen R.: <i>Classical and Object-Oriented Software Engineering with UML</i> , 4th ed., McGraw-Hill, 1998 Schach, Stephen R.: <i>Classical and Object-Oriented Software Engineering</i> , 3rd ed., Richard D. Irwin, 1996 Schach, Stephen R.: <i>Software Engineering</i> , 2nd ed., Richard D. Irwin (formerly Richard D. Irwin/Aksen Associates), 1993	1st ed. 1990 8th ed. 2011 Translated to Chinese, Korean, Spanish
4 Pfleeger, Shari L. and Atlee, Joanne M.: <i>Software Engineering: Theory and Practice</i> , 4th ed., Pearson (formerly Ellis Norwood), 2009 Pfleeger, Shari L.: <i>Software Engineering: Theory and Practice</i> , 2nd ed., Prentice Hall, 2001	1st ed. 1998 4th ed. 2009 International Edition, 4th ed. 2009 Translated to Spanish, Russian
5 van Vliet, Hans: <i>Software Engineering: Principles and Practice</i> , 3rd ed., Wiley, 2010 van Vliet, Hans and Kroese, Stenfort: <i>Programvaruteknik</i> (Software engineering) (in Dutch) [Publisher unknown], 1988	Dutch ed. 1988 1st ed. 1993 3rd ed. 2010 Student Edition, 3rd ed., 2010 Translated to Chinese, English, German, Italian

The first two textbooks, by Sommerville and by Pressman, stand out for their high numbers of editions, 10 and 9, respectively. Interesting, each's first edition was published in 1982, about when SE began to emerge as a profession. Also, around this year, the first courses in SE were introduced in many university computer science programs [33]. While all editions of the Sommerville textbook have only him as the author, the last two editions of the Pressman textbook have Maxim as a co-author. The same happened to the Pfleeger textbook; the last two editions have Atlee as a co-author. Each textbook but the one by van Vliet was published in English first and then translated to other languages.² Fig. 1 shows the number of editions and the number of translations for the five SE textbooks.

2.1.2. Groups and requirements elicitation in SE textbooks

The first result of the review is that the most successful SE textbooks deals, albeit in its own way, with the GvsI issue for SE processes in general. Only a few give detailed numerical group size recommendations for specific activities (e.g., for testing), and none gives a detailed recommendation for requirements elicitation group size.

All the SE textbooks consider the size and complexity of the CBS being developed as key factors impacting the GvsI issue: projects to develop big CBSs need more people than those to develop small CBSs.

² For more details about the textbooks and the results of the questionnaire, the reader is referred to an extended version available online as technical report 31.

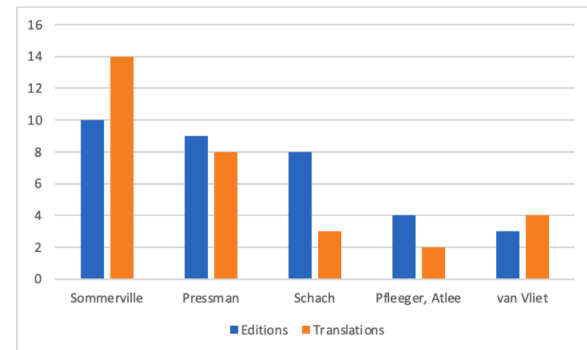


Fig. 1. Numbers of editions and translations of the SE textbooks.

For example, Pressman states that if the CBS to be developed is small, then it is better to have a single individual than a multi-person group developing it. However, none of the textbooks give guidelines on how to quantify the size of a project.

Limits on the duration of a CBS development project, e.g., to beat competitors to the market or to meet the CBS client's deadline, force additional constraints on project staffing. For example, Schach declares that "Today's projects are too large to be completed by a single individual within the given time constraints. Instead, a team of software professionals collaborate on the project".

Another reason in favor of large groups reported in SE textbooks is the need to cover all the skills and expertises related to the CBS project. Hence, a group's size can be no smaller than the number of people needed to provide them.

Each textbook mentioned in Table 1 points to the critical role of communication among group members, underlining that its effectiveness and efficiency depend on the amount of communication overhead in the group. Some of them quantify the communication overhead using the classical quadratic formula for the number of communication lines in the group as a function of the number n of members in the group, $n \times (n-1)/2$.³

Some of the textbooks mention behavioral factors affecting communication effectiveness and efficiency. For example, status differences among group members can affect communication, if status-conscious group members allow managers and the more experienced members to dominate the group's communication. To mitigate the risks arising from communication overhead or lack of domain or professional expertise in groups for SE processes, all five textbooks suggest using creativity techniques [19,42], in particular, brainstorming. Thus, indirectly, group size recommendations for brainstorming – 12 people, both experts and novices, with the latter being expected to provide more innovative ideas [35,36] – become implicit group size recommendations for SE processes. However, recent studies of brainstorming have shown that, depending on the kind of problem to be solved by a brainstorming session, small groups and even individuals may be more effective than large groups [2,27,44]. Guidelines for the classical JAD (Joint application development) technique suggest a group size corresponding to the number of possible stakeholders inside and outside the system ([56]; David [9]). However, the textbooks that suggest using brainstorming and JAD do not mention that there are guidelines for the size of groups for these methods. The Schach textbook discusses Agile development methods and its concept of pair programming [1]. However, it does not

³ The amount of time needed from group members to service each line of communication varies from group to group, and even within a group. No matter what the minimum time per line is for any group, there is always some number m of members such that the amount of time required to service the $m-1$ new lines of communication created by adding the m th group member adds up to more time than the m th member adds to the total work time of the group. Thus, we can say that there is an upper bound on a useful group size.

explicitly address the GvSI issue in this context.

2.2. Requirements engineering textbooks

2.2.1. Choosing the RE textbooks

The second part of the textbook review focused on RE textbooks. Books and the related information were retrieved using the same search process described for the SE textbooks. Then, for filtering RE textbooks, we applied criteria weaker than those for SE books. The differences arose from the necessity to consider that RE is a specialized sub-area of SE and therefore the target audience and market for an RE textbook are smaller. Not only will there be fewer editions, but also the textbook will be translated into fewer languages. In addition, RE as a research and professional area started later than SE, i.e., in the early 1990s. So, the first RE textbooks were published more than 15 years after the first SE textbooks were. Finally, the focus of this paper is on RE, therefore, we needed to include more RE textbooks. The main weakening is that we counted a reprinting of an unchanged edition as well as a new edition as evidence of popularity. We considered RE textbooks that

- have reached at least three editions or reprintings OR (not “AND”)
- have been translated into at least one language other than the original language.

In the end, we found ten qualifying textbooks, and they are listed in Table 2, ordered by decreasing numbers of versions (to simplify the ensuing discussion, we use the term “version” to mean “edition or reprinting”).

The structure of Table 2 is the same as that of Table 1. A first analysis of Table 2 confirms that RE became established as a taught discipline in the second half of the 1990s. The table also shows that each of the main publishers has at least one successful RE textbook. The first editions of the RE textbooks by Sommerville & Sawyer and by Kotonya & Sommerville were published in 1997 and 1998 respectively, about a half decade after RE became a named field. The first editions of the RE textbooks by industrial RE practitioners Robertson & Robertson and Wiegers were published in 1999. Fig. 2 shows the number of editions or reprintings and the number of translations for the ten RE textbooks.

An interesting observation is that eight of the ten textbooks have at least 350 pages in their last versions, with Pohl’s textbook reaching 813 pages. Also, when a textbook has more than one version, the last version has more pages than the first. A high page numbers reflect an in-depth coverage of topics of the textbooks involved. The two exceptions, the textbooks by Hull et al. and by Kotonya & Sommerville, with fewer than 300 pages, cover topics to only the depth necessary to give general information, probably accounting for their success and being listed in the table. On the other hand, a longer textbook does possibly address many specific topics, such as those related to our research question, “What is a suitable size for a requirements elicitation group?”.

2.2.2. Groups and requirements elicitation in RE textbooks

Recall that the assumed scenario for the study is that an analyst needing advice on the GvSI issue for ReqElic activities consults the SE and RE textbooks that he or she used during his or her education. The main result of the RE textbook part of the study is that the analyst will find only generic advice.

Almost every reviewed RE textbook recommends group work as important for conducting good RE. However, there are differences in how work group is introduced. Some textbooks, i.e., those by Wiegers & Beatty, by Widrig, and by Pohl, suggest work group more strongly than others and describe RE specifically in terms of group activities. On the other hand, other textbooks, i.e., by Kotonya & Sommerville, and by Sommerville & Sawyer, rarely mention group work. In addition, the RE textbooks in Table 2 that talks about group work, talks about it for the entire RE process. Very few textbooks describe group work specifically for carrying out ReqElic activities. Among them, the textbook by

Table 2
Requirements engineering textbooks.

	Textbook	Editions
1	Dick, Jeremy, Hull, Elizabeth and Jackson, Ken: <i>Requirements Engineering</i> , 4th ed., Springer, 2017 Hull, Elizabeth, Jackson, Ken and Dick, Jeremy, <i>Requirements Engineering</i> , 3rd ed., Springer, 2011	1st ed. 2002 (213 pages) 4th ed. 2017 (239 pages) Translated to German, Korean
2	Robertson, Suzanne and Robertson James: <i>Mastering the Requirements Process: Getting Requirements Right</i> , 3rd ed., ACM Press/Addison-Wesley, 2012 Robertson, Suzanne and Robertson James: <i>Mastering the Requirements Process</i> , Addison-Wesley, 1999	1st ed. 1999 (404 pages) 3rd ed. 2012 (reprinted in 2014) (547 pages) Translated to Chinese, Japanese
3	Wiegers Karl E. and Beatty, Joe: <i>Software Requirements</i> , 3rd ed., Microsoft Press, 2013 Wiegers Karl E.: <i>Software Requirements</i> , Microsoft Press, 1999	1st ed. 1999 (350 pages) 3rd ed. 2013 (reprinted in 2015) (637 pages) Translated to Chinese, Czech, German, Portuguese
4	Maciaszek, Leszek: <i>Requirements Analysis and Systems Design</i> , 3rd ed. Pearson, 2007 Maciaszek, Leszek: <i>Requirements Analysis and Systems Design. Developing Information Systems with UML</i> , Addison-Wesley, 2001	1st ed. 2001 (378 pages) 3rd ed. 2007 (656 pages) Translated to Chinese
5	Laplace, Phillip A.: <i>Requirements Engineering for Software and Systems</i> , 3rd ed., CRC Press, Taylor & Francis, 2017	1st ed. 2009 (241 pages) 3rd ed. 2017 (400 pages) No translations
6	Leffingwell, Dean and Widrig, Don: <i>Managing Software Requirements. A Use Case Approach</i> , Addison-Wesley Professional, 2003 Leffingwell, Dean and Widrig, Don: <i>Managing Software Requirements. A Unified Approach</i> , Addison-Wesley Professional, 1999	1st ed. 1999 (528 pages) 2nd ed. 2003 (reprinted 2012) (544 pages) No translations
7	van Lamsweerd, Axel: <i>Requirements Engineering: From System Goals to UML Models to Software Specifications</i> , 2nd ed., Wiley, 2011	1st ed. 2009 (682 pages) 2nd ed. 2011 (reprinted in 2013) (682 pages) Translated to German
8	Kotonya, Gerald and Sommerville, Ian: <i>Requirements Engineering: Processes and Techniques</i> , 7th reprinting, Wiley, 2005	Only ed. 1998 (294 pages) 7th reprinting 2005 (294 pages) No translations
9	Sommerville, Ian and Sawyer, Pete: <i>Requirements Engineering: A Good Practice Guide</i> , 6th reprinting, Wiley, 2006	Only ed. 1997 (404 pages) 6th reprinting 2006 (404 pages) Translated to Italian
10	Pohl, Klaus: <i>Requirements Engineering: Fundamentals, Principles and Techniques</i> , Springer, 2010 Pohl, Klaus: <i>Requirements Engineering: Grundlagen, Prinzipien, Techniken</i> , (in German), Dunkt, 2008	German ed. 2008 (741 pages) English ed. 2010 (813 pages)

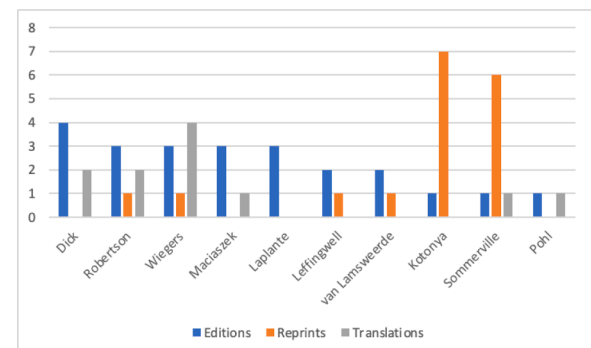


Fig. 2. Numbers of editions or reprintings, and translations of the RE textbooks.

Laplace devotes a whole section to group work in a chapter on requirements elicitation. But in general, only a few RE textbooks actually address the research question and the GvSI issue focusing on

requirements elicitation.

Many of the RE textbooks, i.e., those by Robertson & Robertson, by Wiegers & Beatty, by Laplante, by Leffingwell & Widrig, by Dick et al., and by Sommerville & Sawyer, describe skills for this role. Basically, an analyst has to have both soft skills — the ability to effectively interact inside and outside groups, and to act as a stakeholder representative — and hard skills — the ability to apply SE, RE, and business knowledge. The descriptions of these skills are used to give suggestions for the composition of groups, starting from the basic consideration that any productive group has to include analysts that cover all the named skills.

For group activities in RE, almost every textbook suggests workshops. Some textbooks recommend workshops specifically for requirements elicitation. Wiegers & Beatty suggest requirements workshops; Robertson & Robertson suggest event workshops, and Van Lamsweerde suggests group session in which each participant has defined roles; and Leffingwell & Widrig consider requirements workshops to be the most powerful requirements elicitation technique.

For the application of group techniques, some textbooks provide recommendations on group sizes. Those that suggest workshops, recommend — in different ways and for different reasons — to keep groups small, but to include the right participants. The common goal for these recommendations is to minimize group dynamics [13] and to maximize skill and stakeholder coverage. For example, Dick et al. recommend splitting a large workshop team into small ones, but say nothing about the sizes of these small teams. Pohl suggests 5 to 15 participants to represent all relevant stakeholders and to cover all the types of requirements. He also advises avoiding too-large groups that suffer the negative effects of group-communication overhead by splitting a large group into sub-groups, each for a specific process.

Robertson and Robertson recommend not to add people to a late project, to avoid making it even later; however, they offer no other hint about group sizes. Wiegers & Beatty say that a small team, with 5 to 6 participants, works much faster than a larger team.

For a brainstorming session, Maciaszek suggests having 12 to 20 participants, even though the original description of brainstorming recommends 12 as ideal group size [36]. For a JAD group, he suggests having even more, namely 25 to 30 participants. Leffingwell & Widrig recommend that a group should have 10 to 30 participants, to cover all the necessary roles and activities. However, these recommendations are for groups working in the whole RE process, to cover all roles and activities, and are not specifically for ReqElic groups. Van Lamsweerde says that it is possible to adopt an Agile approach for a CBS project and assign it to a small group, if the CBS to be developed is small enough. However, he does not give specific numbers to quantify either the size of the project or the size of the group.

Finally, Kotonya & Sommerville discuss the size of a team that is conducting requirements review. They say that such a team normally has 4 or 5 members. They also emphasize that there is no ideal size, as it depends on the size of the CBS being developed and on the number of stakeholders.

Many an RE textbook suggests creativity techniques, usually brainstorming, to help elicit requirements. Sommerville and Sawyer suggest brainstorming for the specific purpose of identifying hazards in a CBS; but they add that creativity may not be necessary for this specific kind of brainstorming, confirming that many authors use the term “brainstorming” as a synonym of meeting and not as the name of a full-fledged technique, based on a few clear principles and guidelines. Many an RE textbook suggests group techniques other than brainstorming for requirements elicitation, usually JAD.

Altogether, the conclusion of the first study, is that none of the textbooks in Table 1 and Table 2 provides real help, based on empirical data, in making decisions about the GvSI issue and about the size of a group for ReqElic. The second study, based on an online survey aims at filling in this gap.

3. The online survey

3.1. Design and realization of the survey

To investigate the research question defined for our study, we designed a survey based on an online questionnaire.

In designing it, we considered general guidelines and good practices for questionnaires [32,48]. We designed a pilot questionnaire, and we gave it to three practicing analysts to validate that the questions were understood as we intended them to be. We received and implemented suggestions to improve some of the questions, simplifying the wording and checking the use of the same words for a given concept. In particular, we abbreviated “requirements elicitation” by “ReqElic”, to distinguish it from “RE”; and “business or requirements analyst” as “BoRA”. Finally, we decided to provide an incentive to answering the questionnaire by offering each respondent an opportunity to win a copy of some RE books.

The final questionnaire has 12 questions, Q1–Q12, each of which has from one to four subquestions, for a total of 19 subquestions to answer (<https://zenodo.org/record/7616991>). Overall, the questions ask each respondent about how ReqElic is done in CBS development projects at his or her company.

Questions from Q1 to Q6 are multiple-choice, with 4 possible answers, “in all”, “most”, “some”, “none” of the aspects of the project specified in each question.

Q7 and Q8 investigate the usual and ideal group size in the respondent’s company, respectively.

Questions from Q9 to Q11 are open questions whose goal is to investigate the reasons a respondent suggested a given ideal group size (Q9) and factors to recommend or not to recommend groups of BoRAs (Business or Requirements Analysts) for ReqElic (Q10 and Q11).

The last question, Q12, asks to list three keywords that characterize the domain or sector of the respondent’s company’s software system development projects.

3.2. Results of the questionnaire

In the end, in late August 2012, we deployed at GoogleDocs, now GoogleForms, an online questionnaire titled “Requirements elicitation (ReqElic) in my company”. To implement the incentive to answer the questionnaire, we added a last question, inviting the respondent to be entered into a drawing to win a copy of the Proc. next REFSQ (Requirements Engineering: Foundation for Software quality) Conf. We began to solicit respondents through a variety of venues. We sent an advertisement describing the questionnaire to analysts or software development managers that we knew and asked that they send the advertisement on to other people in similar roles. We posted the advertisement and the propagation request:

- on the Facebook, Google+, LinkedIn, ResearchGate, Slideshare, and Twitter accounts of one of the authors,
- on several e-mail lists, e.g., IIBA, INCOSE, Requirements Engineering Network’s forum, RE-online, and Yahoo’s Requirements- Engineering Group,
- at several LinkedIn groups, including AICA, Community of Practice Systems Engineering (CoP SE), America’s Requirements Engineering Association, Business Analysts - Bangalore, ICT Africa, ICT Australia, IEEE Computer Society Italy Chapter, INCOSE, IREB Certified Professional for Requirements Engineering (CPRE), Modern-Analyst.com - Business Analyst Community, Requirements Engineering Specialist Group (RESG), Systems Engineers, and Requirements Engineering.

Thus, we have a convenience-enhanced-by-snowballing sampling. From the first advertising round, we got 53 answered questionnaires and used preliminary findings to corroborate the results of an experiment

about group size for requirements idea generation [44]. During a second advertising round, we sent reminders to the groups and communities of the first round, and we sent the announcement to some new target groups and Conf. attendee lists. After this round, we reached a total of 92 responses and published a descriptive analysis of the answers to the first 8 questions [29].

We cannot know the response rate because we have no idea how many people saw the advertised invitation to fill out the questionnaire. However, our analysis of the answers to Q12, which asks a respondent for three keywords to characterize the domain or sector of his or her company's software system development projects, allows us to be confident that the respondents were working in many different companies or in at least different divisions of the few large multi-domain companies [30]. The following sub-sections describe the results of the analysis of the answers to the questionnaire.

3.2.1. Q1 — My role in ReqElic

Q1 investigates how respondents are involved in ReqElic, asking the frequency with which they play roles in the following subquestions:

- I'm directly involved as a business or requirements analyst (BoRA),
- I'm directly involved as a software engineer,
- I'm supervising ReqElic as a project manager or similar role, and
- I'm involved in ReqElic as a representative of the client or customer.

Results show that almost all the respondents are the kind people the survey was targeting: 93% of them play the role of BoRA in at least some of the projects (Fig. 3). This result is relevant to establish the reliability of the questionnaire's answers. Besides, answers to Q1 highlights that many respondents play multiple roles, an indication that they are working in multiple steps of any CBS development lifecycle, and that, in turn, the size of the group necessary to provide all the needed domain knowledge and expertise can be smaller [37].

3.2.2. Q2 — Group vs. individual activity in ReqElic

Q2 is the most important question for the GvSI issue. It determines the frequency with which requirements identification is done in the respondents' companies in each of three different configurations:

- Requirements are identified as an individual activity, by a single BoRA, working alone
- Requirements are identified as an individual activity, by more than one BoRA, each working separately
- Requirements are identified as a group activity.

Fig. 4 shows that groups are frequently adopted in companies, as only 15% of them do not adopt groups in any of its project. However, in about 2/3 of the companies, requirements are identified — in at least some of their projects — as individual activity, i.e., by BoRAs working separately (68% of the companies) and by a single BoRA (64% of the

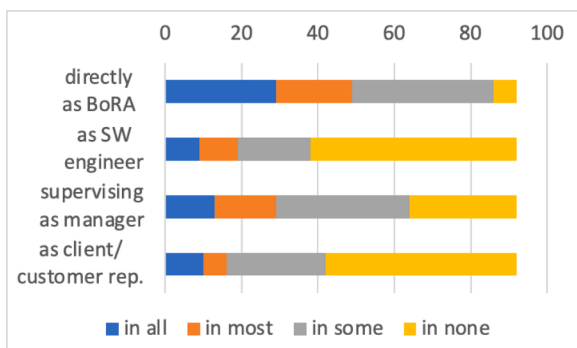


Fig. 3. Q1 My role in ReqElic.

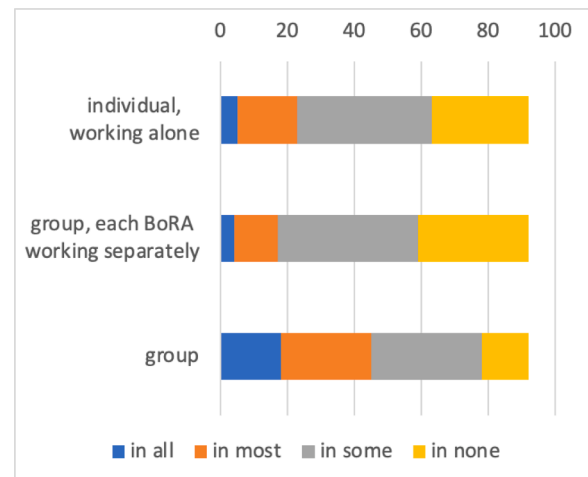


Fig. 4. Q2 Group vs individual activity in ReqElic.

companies).

3.2.3. Q3, Q4, Q5 — Group activity in ReqElic as a function of the size, deadline, and complexity of the project

Questions from Q3 to Q5 determine the frequency with which requirements identification is done in the respondents' companies as a group activity for three different kinds of projects:

Q3: Requirements are identified as a group activity in large projects.

Q4: Requirements are identified as a group activity in tight-deadline projects.

Q5: Requirements are identified as a group activity in complex (innovative or multi-disciplinary) projects.

The answers to Q3, Q4, and Q5 (Fig. 5), show that requirements are identified in groups more frequently, in large and complex projects (90%), than in tight-deadline projects (82%).

According to these results, the size and the complexity of a project are more important than time constraints, as indicated by “tight-deadline”, in choosing groups for ReqElic.

3.2.4. Q3, Q4, Q5 crossed with each other — Group activity in ReqElic vs. different combinations of the characteristics of the project

To explore requirements identification as a group activity in large, tight-deadline, and complex projects more deeply, we conducted pairwise crossings among Q3, Q4, and Q5. Figs. 6, 7, and 8 show the results of the crossings. These graphs show that there is a strong association between how often requirements identification is done in groups in different kinds of projects. The more requirements identification is a

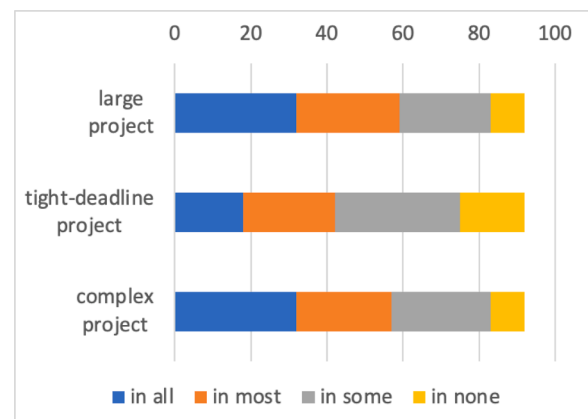


Fig. 5. Q3, Q4, Q5 Group activity in ReqElic as a function of the characteristics of the project.

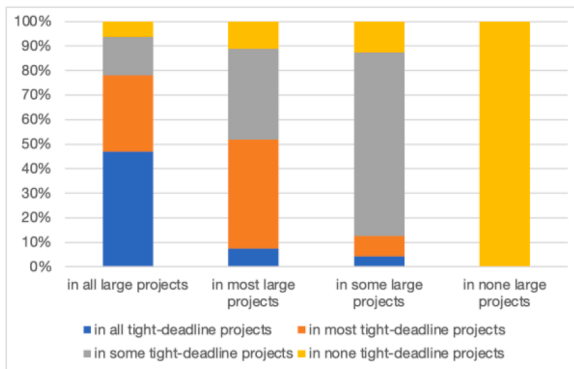


Fig. 6. Q3vs Q4 Group activity in ReqElic for large vs tight deadline projects.

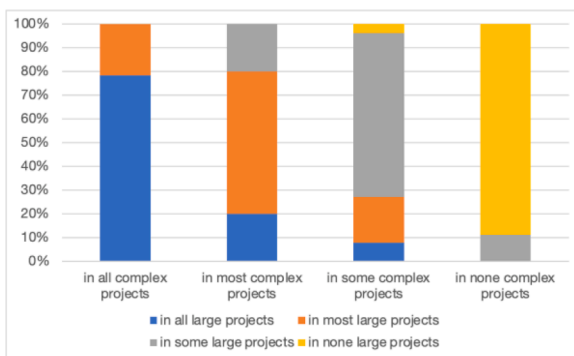


Fig. 7. Q3 vs Q5 Group activity in ReqElic for large vs complex projects.

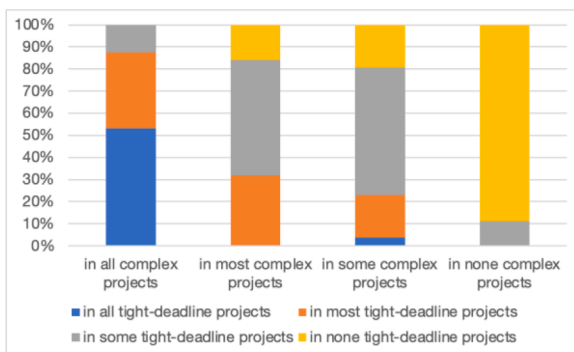


Fig. 8. Q4 vs Q5 Group activity in ReqElic for tight-deadline vs complex projects.

group activity in one type of project, the more requirements identification is a group activity in other types of projects. The leftmost bars of Figs. 6, 7 and 8 highlight that companies that do requirements identification as a group activity in all and most of their projects for a given type of project, do so more frequently for large-complex projects than for tight-deadline projects.

Crossing the results for the three questions investigating group requirements elicitation vs. the characteristics of the project, highlights that groups are adopted more often for projects that are both large and complex than for the other combinations, i.e., large or complex and tight-deadline projects.

3.2.5. Q6 — The kind of ReqElic techniques used in ReqElic group activity

The subquestions of Q6 determine the frequency with which the ReqElic specified techniques is used in ReqElic group activity sessions in the respondents' companies:

- We use JAD in group ReqElic sessions
- We use brainstorming in group ReqElic sessions
- We use other creativity techniques in group ReqElic sessions.

Fig. 9 shows that about two times as many respondents are using brainstorming or other creativity techniques than JAD in group ReqElic sessions in at least some of the projects, and in fact, in about 80% of the projects. A result that suggests companies use creativity techniques for requirements idea generation. Increased creativity or something similar is cited in many of the explanations given as answers to Q9, for preferring the use of groups instead of individuals as ideal sizes and in many of the factors given as answers to Q10, to recommend using groups of BoRAs for ReqElic.

3.2.6. Q7 – Size of the groups

Q7, "Groups usually consist of", asks the respondents to choose one of five different sizes, measured in numbers of BoRAs, ranging from 2 to >5 BoRAs (Fig. 10). The answers to Q7, show that in the respondents' companies, the usual group sizes, in order of decreasing occurrence, are 2, 3, 4, and >4 (grouping the answers for sizes 5 and >5). Companies adopt different usual group sizes, but 68% of the respondents say that the usual group size is one of the two smaller group sizes, 2 or 3.

3.2.7. Q8 – Size of ideal groups

The question on ideal size was formulated in such a way to have respondents chosen between three possibly way to distribute four analysts, to get useful data about the Gvsl issue. Q8 determines ideal group sizes in the respondents' companies, for each of three different configurations,

- 4 individuals
- 2 groups of 2
- 1 group of 4.

Fig. 11 shows that having 4 analysts, "2 groups of 2", are preferred to "1 group of 4", and to "4 individuals". That is, groups are preferred to individuals, and among groups, smaller groups are preferred. In more detail, among the respondents, 42% chose 2 groups of 2, 37% chose 1 group of 4, and 21% chose 4 individuals. These results confirm that there is not 1 ideal size for REqElic sessions in companies, and that 1 in 5 of the respondents suggest having individual instead of groups.

3.2.8. Q7 Crossed with Q8 – Size of groups vs size of ideal groups

The results of crossing the answers to Q7, about usual group sizes, with the answers to Q8, about ideal group sizes (Fig. 12), show that those whose usual group size is "2 BoRAs" tend to select an ideal group size of "2 groups of 2", and similarly, those whose usual group size is "5 BoRAs" or ">5 BoRAs" tend to select an ideal group size of "1 group of 4". Those whose usual group size is "3 BoRAs" or "4 BoRAs" tend to select ideal group sizes of "2 groups of 2" and of "1 group of 4" equally

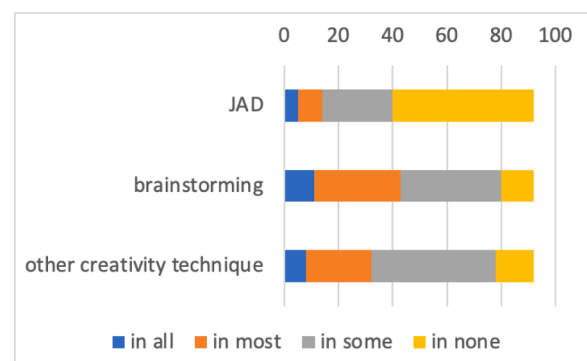


Fig. 9. Q6 The kind of ReqElic techniques used in ReqElic group activity.

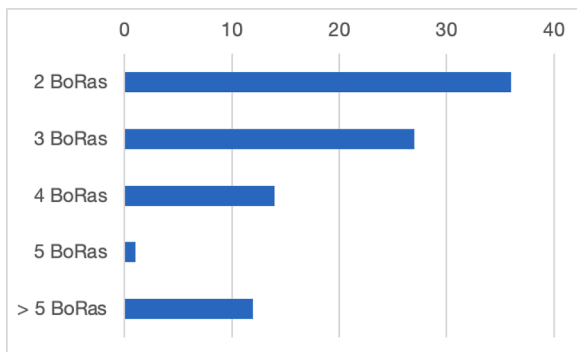


Fig. 10. Q7 Usual size of the groups.

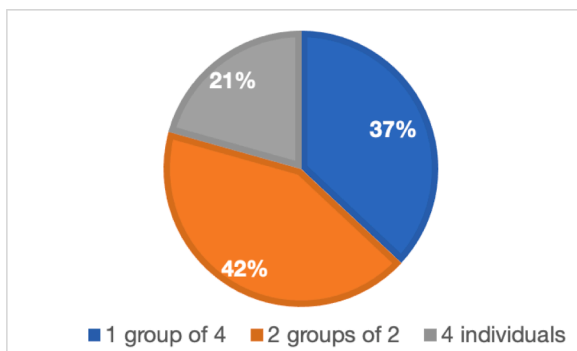


Fig. 11. Q8 Size of ideal groups.

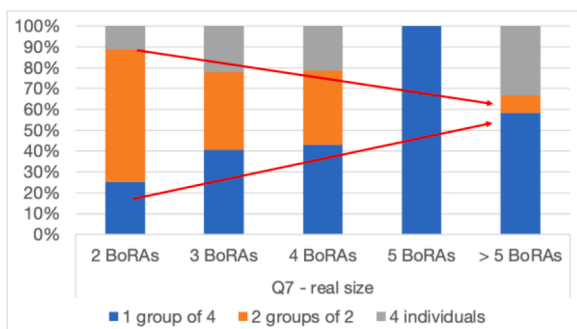


Fig. 12. Q7 vs Q8 Size of groups vs Size of ideal groups.

often.

In a nutshell, many, but not all, of the respondents suggest as the ideal group size one that is similar to the usual size. Reasons for these differences are investigated in the following questions.

3.2.9. Q2 Crossed with Q8 – Group vs. individual activity in ReqElic vs size of ideal group

Fig. 13, which crosses the answers to Q2 (Group vs. individual activity in ReqElic) with the answers to Q8 (Size of ideal group), shows that respondents that

- 1) work more frequently in groups in all their projects, and therefore,
- 2) work more frequently in other configurations in none of their projects,
- 3) tend to choose “4 individuals” least often as their ideal group size, and
- 4) tend to choose “1 group of 4” most often as their ideal group size.

The explanations these respondents give in Q9 to justify their

answers about the size of ideal group, mention both (1) problems related with individual work and (2) advantages of group work, and they say, for example:

- “Without working together on the req base, there’s going to be lots of inconsistencies and gaps.”
- “When 4 individuals are involved in the same project, they must cooperate.”
- “I think a group up to 4 members is more creative than one alone. In discussions the members have more new ideas than thinking alone.”

“Each person contributions can build on the others. On the other hand, the respondents who say that they work in companies in which requirements are identified as an individual activity by more than one BoRA, each working separately, do not chose “1 group of 4” as their ideal group size; instead, they chose “2 groups of 2” as their ideal group size more often than “4 individuals”. A deeper analysis of the open questions (Sections 3.3.11, 3.3.12) shows the reasons why some people would prefer to work in a configuration that is different from the configuration in which they actually work.

3.2.10. Q3, Q4, Q5 Crossed with Q7 – Group activity in ReqElic as a function of the size, deadline, and complexity of the project vs size of group

Fig. 14 shows the result of crossing Q3, Q4, Q5 about how often requirements identification is done in groups, in large, tight-deadline, and complex projects, respectively, with Q7, about usual group sizes.

The main result is that, regardless of project type, the less often requirements identification is a group activity in a company, the more often the usual group size is “2 BoRAs”. Conversely, the more often requirements identification is a group activity in a company, the more often the usual group size is one of the larger numbers of BoRAs.

3.2.11. Q9 – Explanation to answer about the size of ideal groups

Q9 “Here are my explanation to answer to the previous question, about the size of ideal groups” is an open question. The answers were very different from each other, both in length and in structure. Problems in analyzing their content are those typical of interpreting sentences in natural language: it is necessary to deal with sentences fragments, synonyms, jargon, etc.

The analysis was carried out in two passes. In the first pass, authors Mich and Sakhnini worked together to identify the different aspects or concepts cited in the answers using different terms. In this pass, the two authors worked together iteratively to examine ten answers in detail. First, they classified answers according to their mentioned concepts. For example, for the answer “sufficient time, budget allows, new projects (including lack of experience in domain)”, the result is “sufficient time”, “budget allows”, “new projects”, “lack of experience in domain”.

Expressions citing similar aspects – factors from now on – are grouped, so that for example, “increased cost associated with additional personnel” was grouped together with “budget allows”; similarly, “every person has different skills” was grouped with “A group working as a team as the best mix of skills and experience to the table”. In a final step, authors used these pre-processed expressions to generalize them. For example, “budget allows”, “increased cost associated”, “sufficient time” were recognized as factors characterizing a project category. Expressions that cite experience or skill were recognized as characterizing a *People* category. In this way, the two authors identified 4 categories according to the involved concepts:

Relations, including factors related to interactions between group members, described in terms of communication, coordination, collaboration, responsibility, and viewpoints.

People, including factors characterizing an analyst as an individual,

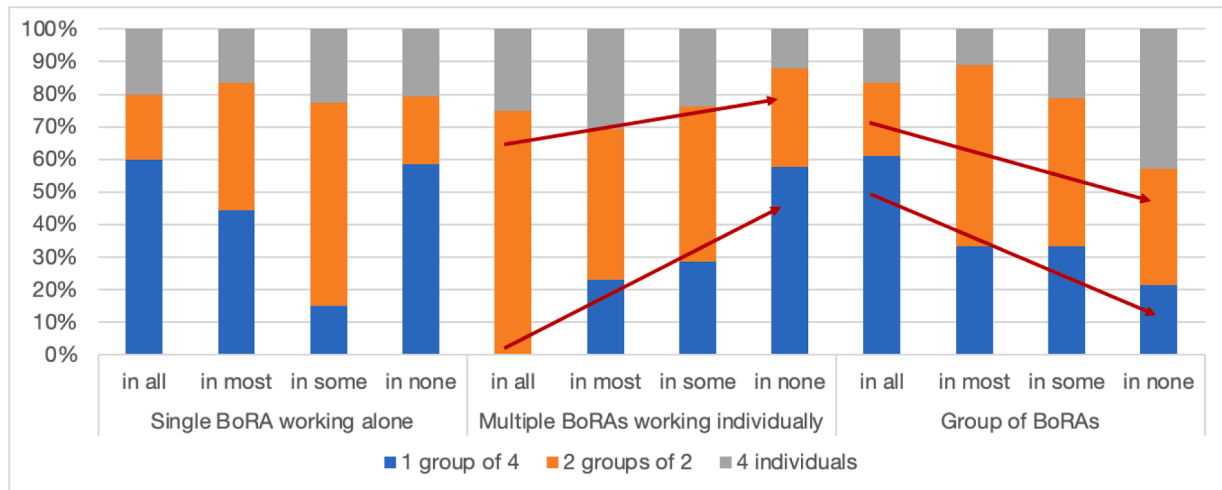


Fig. 13. Q2 vs Q8 Group vs. individual activity in ReqElic vs Size of ideal group.

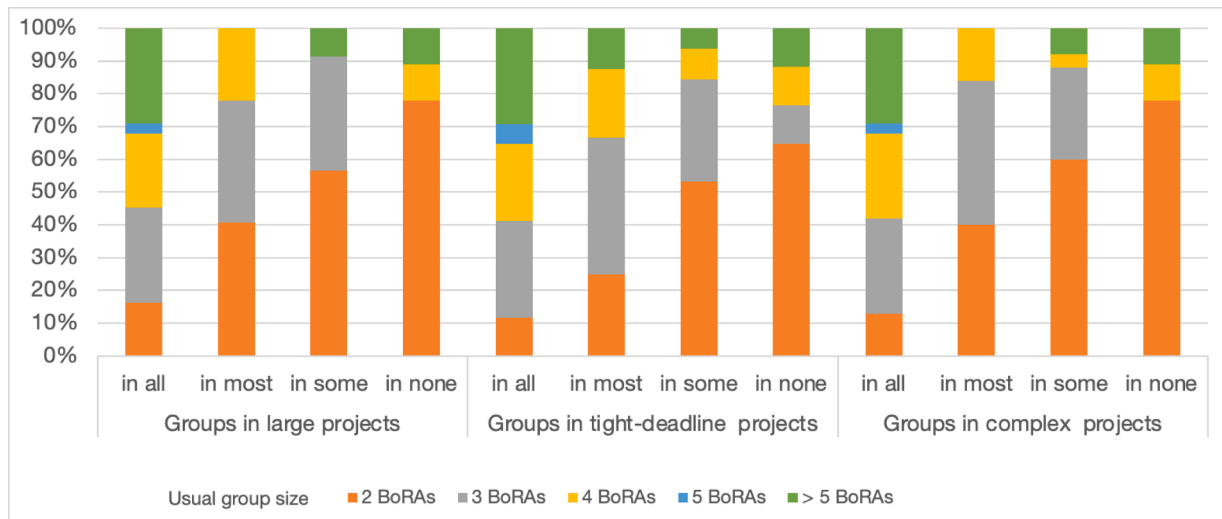


Fig. 14. Q3, Q4, Q5 vs Q7 Group activity in ReqElic as a function of the size, deadline, and complexity of the project vs Size of group.

for example, by his or her expertise and self-confidence.⁴

Project, including all factors related to project management, e.g., project size, timing, and complexity, and available human resources and other costs.

Output, including factors used to describe the expected results of requirements elicitation, often described in terms of qualities of requirements specifications, e.g., better coverage, fewer duplicate, not conflicting requirements.

In the second pass, the first author classified each answer into one or more of the four categories, according to all the factors that appeared in it. Fig. 15 summarizes the answers to Q9 by showing the number of times factors of each category appeared among the answers.⁵ The categories of factors mentioned in participants' answers to Q9 were, in order of decreasing frequency, "People", "Relations", "Output", and "Project".

Reasons given for the choice of the ideal size, grouped by category,

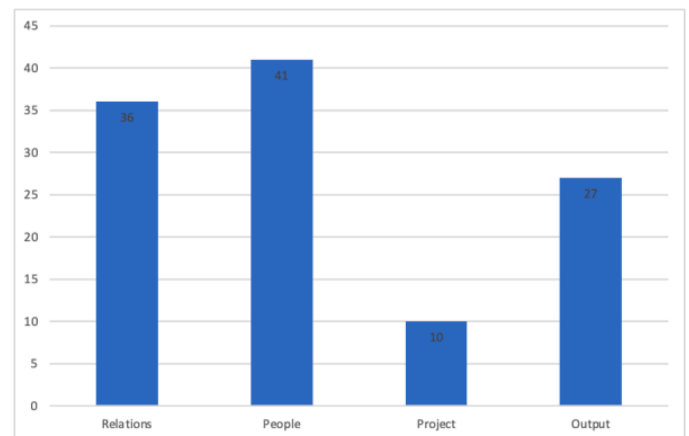


Fig. 15. Q9 Reasons for the choice of the ideal size.

⁴ For a detailed explanation of the relational and individual characteristics see [54].

⁵ Note that ten respondents did not answer Q9 at all, and many answers mentioned factors in more than one category. Hence, the sum of the lengths of the bars is not expected to be 92, the number of respondents.

are as follows:

People: People factors such as "expertise in the project domain" and "expertise in SE methods and tools" are considered to have positive effects on group configurations. The people factors, "self-confidence" and

“responsibility” are considered improved by individual configurations.

Relations: The individual and the small-group configurations are considered ways to reduce conflicts. On the other hand, relation factors are considered to have a positive impact by those that suggest the large-group configuration. Good collaboration is considered to have a positive effect on a group’s ability to reach better solutions, and it allows the presence in a group of more stakeholders to present more viewpoints. In fact, these human factors, both individual and relational, are the most frequently cited reasons for respondents’ answers.

Output: Group configurations help avoid the negative output factors of “duplicates” and “inconsistencies”. Individual configurations help achieve the positive output factor of “more original ideas”. The “productivity” positive output factor improves in the small-group configurations only if conflicts are avoided.

Project: While project factors were rarely mentioned in the answers, some answers did indicate that it’s essential to have more than four in a group that is building a large, complex, or interdisciplinary system.

3.2.12. Q10, Q11 – Factors to recommend and to not recommend using groups

Q10 and Q11 ask the respondent to list three factors that would cause him or her to recommend or to recommend *not* using groups for ReqElic, respectively. The answers to Q10 and Q11 were classified by the categories of factors identified for Q9, and the results of this classification are shown in Figs. 16. Since 78% of the respondents chose a group configuration as their ideal group size, it is no surprise that Fig. 16, giving reasons for his or her recommending using groups of for ReqElic (blue bars), is similar to Fig. 15.

In fact, some respondents answered Q11, asking three reasons to not recommend using groups in ReqElic, by saying that there is no reason to recommend not using groups of BoRAs for ReqElic. For each of the factors there are advantages and disadvantages, so that factors in any one category can be offered both for and against using groups of BoRAs for ReqElic, in kind of an antisymmetric way. Project factors are the least frequently mentioned in recommending using groups, but are the most frequently mentioned in recommending not using groups for ReqElic. However, output and people factors are more frequently mentioned in recommending using groups, but are less frequently mentioned in recommending not using groups for ReqElic. Finally, relations factors are mentioned with the same frequency in recommending either using or not using groups in ReqElic.

Table 3 gives some examples of quotations from respondents’ answers, grouped by the categories into which they were classified; by whether or not they recommended using groups for ReqElic.

The results of the study, in particular those concerning factors to be considered when deciding whether to form ReqElic groups are integrated with the results of the textbooks reviews in Section 4.

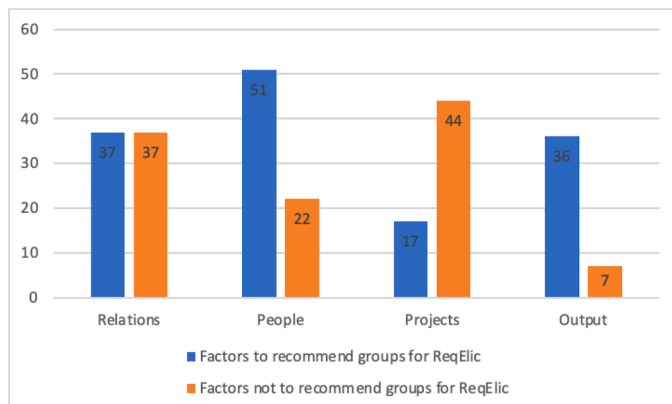


Fig. 16. Q10-Q11 Factors to recommend and not to groups for ReqElic.

3.2.13. Q12 – Keywords characterizing the domain or sector of the projects

Q12 asks the respondent to list three keywords that characterize the domain or sector of his or her company’s software system development projects. Unfortunately, many a respondent answered with information on the types of systems he or she worked on, and not the domain or sector, per se. So, we followed a two-pass coding procedure similar to that we applied for the open questions to identify three main types of projects:

- Information systems, e.g., Enterprise Resource Planning systems
- Software systems, e.g., embedded systems
- Web sites, e.g., e-learning sites, e-commerce sites, portals, recommendation systems.

Results highlighted that some companies work on more than one type of projects, with 58, 35 and 26 for each type respectively. A crossing of the results for the type of systems and the core questions, Q7, Q8 showed not significant association, possibly requiring a finer classification of the systems.

4. Discussion of the results of the two studies

Sections 2.1 and 2.2 review the most successful SE and RE texts – the ones most likely to be on the bookshelves of an analyst – with respect to how they treat the Gvsl issue for ReqElic, i.e., whether requirements elicitation should be carried out by a group or by an individual, and if by a group, how big is the group.

The SE and RE textbooks agree on the *need for group work* for CBS development activities in general, but usually fail to consider the specific ReqElic activity. Many a textbook says that groups are useful for large projects, to have the necessary expertise, or to get new ideas or requirements. Group dynamics are often considered as risks that could reduce the efficiency and effectiveness of group work. Therefore, anyone staffing a group has to consider the tradeoff between (1) having the right skills and roles, which is more easily achieved with larger groups, and (2) mitigating group overhead, which is worse with larger groups.

Many textbooks give recommendations for *sizes of groups*, but such recommendations are usually described without referring to specific steps of a CBS project. Few give suggestions concerning group sizes specifically for ReqElic activities. Many of the textbooks include a description of some ReqElic techniques, but only a few provide information about the number of people to involve in performing them.

Guidelines for the size of a group had possibly to be inferred looking for information on the technique in other sources and this is the case also for creativity techniques. The need of creativity techniques to support requirements elicitation and idea generation is recurrent in the RE reviewed textbooks. Brainstorming is suggested by many of them, however, the word “brainstorming” is often used in the current parlance, as a synonym of “meeting”, when it is used as a noun, or of “inventing”, when it is used as a verb.

Finally, it is important to remark that none of the numbers given by the SE and RE textbooks for groups size appears to have a formal empirical basis; that is none is described as arising from an empirical study. Each seems to arise from the personal, possibly industrial experience or knowledge of its textbook’s authors.

Results of the second study represent a contribution towards bridging this gap. The analysis of the answers to the 12 questions in the questionnaire allows illuminating some specific aspects of the Gvsl issue in requirements elicitation. First of all, answers to Q1 show that respondents play roles relevant for the goal of the Gvsl issue. Roles, including that of BoRA, related to ReqElic are adopted in the respondents’ companies while the role of BoRA was not explicitly named in the SE textbooks.

Answers to Q2, confirm that there are different choices as regards the adoption of groups for ReqElic. Groups are used for ReqElic, but there are projects in which individual BoRAs are used for ReqElic. This fact is

Table 3

Example quotations from respondent answers that are relevant to categories.

Factors related to	Mention	Examples of quotations from respondent answers that recommend $X = [\text{using} \text{not using}]$ groups of BoRAs for ReqElic X	Quotation with a possible reason for recommendation
Project	characteristics of the respondents' projects and the costs and benefits of having many people	using	• "sufficient time, budget allows, new projects (including lack of experience in domain)" • "small enhancements/bug fixes, small project size, budget disallows"
Output	the quality of the requirements, often in terms of the coverage of different stakeholders needs or requirement details	using	• "increased cost associated with additional personnel" • "4 eyes see more details than 2." • "2 groups are good for X-checking each other's work."
		not using	• "It can be very difficult to reach conclusions which are coherent." • "frustrated with team member output; maybe we expect more but the team member gives a low quality output" • "The group sessions are tiresome, slow, unfocused. Individual requirements elicitation is much more productive and can be reviewed later by the group members."
People	having people with the right skills and expertise	using	• "A group working as a team brings the best mix of skills and experience to the table." • "Every person has different skills." • "They are professionals who have the necessary skills and competencies to effectively manage the requirements" • "elicitation process as well as fulfilling other tasks or activities required." • "Interesting ... If you are working as a group, you really need to have the right mix of skills and experiences." • "No single individual has sufficient breadth and depth of knowledge to develop a full set of unambiguous requirements." • "No one knows all there is to know about a product or service. Groups are the only solution." • "knowledge-intensive and complex environments with only one analyst with necessary skills"
Relations	communication, collaboration, responsibility, agreement, debates, and personality	not using	• When "there is a person with good overall picture" • "Team collaboration is important." • "Collaboration works." • "Collaboration encourages competition." • Groups support "confrontation, cooperation and better analytical phase through explicit verbalization". • Groups are useful to "provide support on difficult or dysfunctional teams (members)". • "to increase the knowledge of the dynamics between people" • For a "collective responsibility for success or failure"
		using	• "Putting people together as a team does not mean you have a team." • "communication overhead" • "interpersonal issues" • "the more people involved the more communication challenges" • "too much chance of miscommunication" • "Multiple people can end up in endless debates." • "Too many BoRAs can go in conflict each other." • "BoRAs tend to have strong personalities (in my experience), a group or BoRAs can be intimidating for the customer." • "In some cases, groups are in disagreement." • "lots of disagreements to work through, possibility that opposing views may slow down the work" • "... analysis paralysis if no one can make decisions on group disagreements. If there's an internal conflict between different business goals, then a team may get sucked into that conflict." • "group dynamics: Unwillingness of analyst to work in the attributed roles, risk the meeting gets out of control because every analyst wants to be involved possibility of creating a leader who makes decisions independently."
		not using	• because of the communication and relational problems and personality conflicts that arise when people work in groups • because of the negative impact of disagreement within and between groups

reported in SE and RE textbooks, usually in connection with a discussion of techniques available to support ReqElic.

According to the results of Q3 to Q5, large, complex, and to a lower extent tight-deadline projects tend to adopt group ReqElic. Similarity of results for large and complex projects are possibly because those are intrinsic characteristics of a project, while time constraints may depend on external factors. In RE and SE textbooks, however, the large size of a project is the most often cited reason for forming groups, while the complexity of a project is rarely explicitly mentioned as a reason for forming groups. They usually say that for a complex project, a large group is needed to cover all the necessary expertises needed. However, recommendations found in textbooks are given for an entire project, and not specifically for ReqElic. Interestingly, two of the answers to Q10, about the reasons to recommend using groups for ReqElic, explain that groups are formed for an entire project, not considering at all ReqElic as a specific step that might require its own skills. Brainstorming and other

creativity techniques are adopted in many companies as group ReqElic techniques (Q6). These results are consistent with the suggestions given in the most recent SE textbooks and in the RE textbooks.

The answers to Q7, about usual group sizes, show that companies adopt different group sizes, and 68% of the respondents say that the usual group size is one of the two smaller sizes, 2 or 3. Answers to the open questions confirm that companies use small groups for ReqElic whenever a project's characteristics permit a small group size. But there are projects for which companies would actually prefer to have larger groups, but they cannot afford the related costs, and they cannot always find the right person for a needed role, so they settle for small groups.

Thus, when there are four individuals available to do a ReqElic activity, the answers to Q8, about ideal group sizes, show that respondents suggest forming two groups of two more often than one group of four, and still more often than to leave the four individuals to do the activity alone and separately (Fig. 11).

The results of crossing the answers to Q7 with the answers to Q8, (Fig. 12), show that if there are many respondents confirming as ideal size the same or the closest to that adopted in their company, there are many that would suggest a different configuration. Answers to the open questions allows identifying factors the respondents consider determinant for the GvSI issue.

The answers to question Q9 (reasons for the choice of the ideal size) to Q10 and Q11 (factors to recommend or not groups for ReqElic) allow formulating some guidelines as regards factors to be considered in forming ReqElic groups, and how to choose the group size for a given project. The categories of factors identified in the reasons given to recommend using or not using groups of BoRAs in ReqElic are Relations, People, Project, and Output.

Interestingly, many of the terms used by the respondents, and in turn to classify their answers into the 4 categories, are also used in the descriptions of group work in the textbooks reviewed in Section 2. Relations and People factors are usually cited in SE and RE textbooks to support the recommendation to keep groups small, i.e., to be able to trade off between reducing the number of people to minimize the number of relationships and increasing the number of people to maximize the coverage of needed skills.

The relevance of Projects factors is emphasized by many authors in terms of size or complexity of the project considered as relevant in deciding the GvSI issue.

Output factors are less frequently discussed in relation to the GvSI issue. It is more often found that groups are necessary for requirements elicitation for large or complex project than to produce quality requirements.

Costa et al., in a recent study investigating the state of the practice in group formation in SE, identified two categories of factors relevant to group formation [8]: technical and non-technical. They emphasize that the technical factors are considered more often, but they are often not enough to allow making a good GvSI decision. Given that non-technical factors are concerned mainly with relations, Costa et al.'s results are consistent with those of this paper.

In general, the explanations given by the respondents in Q9 through Q11 allow saying that factors related to Relations can be equally relevant to recommend and not to recommend groups. People and Output factors are more frequently cited to recommend groups than not to recommend them, whereas the opposite is true for Projects factors (Table 4). These results confirm that the GvSI issue in ReqElic is a multifaceted problem, that needs to be addressed with factors in each category in mind.

5. Threats to validity

To define a feasible scope for the SE and RE textbook reviews, we looked for the most successful ones. A first threat is related to the choice of the textbooks to review. The identification of the textbooks is based on two parameters, the number of editions (versions for RE) and the number of languages they have been translated into. Errors in gathering these data could have been impacted on the lists used for the review in Tables 1 and 2. In addition, such data were last checked in 2021; in the meantime, new editions or versions of the included textbooks could have been published. However, both new edition and translation in a new

language are not very frequent. For both lists, in Tables 1 and 2, we have recently checked data for the excluded textbooks, and none of them would be now included. As for the replicability of the identification of the most successful textbook, checking if there are relevant updates would also require dealing with changes in all the platforms used for the search, whose interfaces have been deeply revised.

A second, more critical problem is that being the GvSI related with a variety of aspects, described using terms that evolved in time, addressed in many different ways by the authors, it is possible that not all the relevant information has been found. Besides, all the analyzed textbooks have hundreds of pages. The availability of online editions of the textbooks allowed mitigating this risk, by using search functions.

For the survey described in Section 3, there are two main potential limitations to be considered. The first is the limited number of respondents, and the second is the long time elapsed between the date on which the questionnaire was made available online and the date on which the 92nd, and last, respondent had filled out the questionnaire.

With 92 filled-in questionnaires, the study represents an exploratory study of the GvSI issue in ReqElic, and its results cannot be considered representative of the population. First, it is impossible to know the sizes of the populations that the 92 respondents and their companies represent, and thus, how representative the respondents and their companies are. Second, we use a variety of tools and online platforms to promote the questionnaire (Section 3.2), so, it is impossible to know the number of analysts that saw it. In turn, it is impossible to calculate any response rate.

A comparison with similar online studies can give ideas of both (1) how typical a response of 92 is and (2) the difficulties of having a larger number of respondents. An online survey focused on SE practices – a wider scope than GvSI in ReqElic – in 2003 got 194 completed responses [34]; a follow-up survey, for a master thesis, got interactions – an interaction is an entry into one or more data fields – from 93 respondents [26]. A broader survey – including international participants – by two of the authors of the 2003 survey, got 119 respondents [21]. Thus, the number of respondents to our questionnaire is in the same ballpark as that to other similar SE surveys.

The long-elapsed time of the study is another threat to the validity of the conclusions as the world changes over time. The long-elapsed time is a direct result of our effort to achieve at least 100 filled-in questionnaires. In this effort, we left the questionnaire online from Autumn 2012 until Spring 2015. During that time, we actively promoted the questionnaire, but we never quite achieved the goal of 100 filled-in questionnaires, receiving only 92, the last of which arrived in January 2015. The effect of the long duration is partially mitigated by the observation that the frequency structure – distribution – pattern of the answers for the first 53 is similar to that of the full 92. We found that the order, among the first 53 respondents, and all 92 respondents, of the chosen frequency answer from most chosen to least chosen, is the same for almost all the questions. Table 5 give as an example, the data for Q1.

In addition, answers to the question about the role a respondent plays in ReqElic, confirm the 92 respondents are the kind of people the survey was targeting. A question whose formulation is critical is Q8, about the ideal size of a group. Due to its formulation, the answers to Q8 cannot be easily compared with those of Q7, about the usual size of a group. However, the choice of the possible answers is prompted by the need to have for Q8 data comparable to those for the same kind of decision about 3 different ways to distribute 4 analysts, and to give the

Table 4
Recommendations of factor categories.

Using Groups	Categories Relevant to Recommending	
	for ReqElic	Not Using Groups
People		Project
Relations		Relations
Output		People
Project		Output

Table 5
Frequency for answer to Q1.

	53 respondents	92 respondents
some of our projects	45%	40%
all our projects	26%	32%
most of our project	19%	22%
none of our projects	9%	7%

same context as for Q9.

The issue for each open question, Q9–Q12, is whether the 92 answers received contain all possible answers that one might see in any larger collection answers, i.e., whether *thematic saturation* has been reached with the 92 answers received. Recent studies that investigated the *saturation point* – the number of data that are necessary to be certain that no new results or themes will emerge if more data were to be gathered – indicate that 92 answers may be enough to have reached saturation. Weller et al. have determined that this number is 50 for many sectors [55].

Tran et al. have shown that in each doubling of the sample size, the number of new emergent themes got geometrically smaller.

Specifically, with the samples of 25, 50, 100, and 200 data, the numbers of emergent themes were 93, 107, 114, and 116 respectively, so that, numbers of new emergent themes emerging in the doublings were 14, 7, and 2, respectively [52]. These results indicate that in the 92 filled-in questionnaires, we probably have enough data to achieve thematic saturation in the open questions. In particular, the sets of reasons given as the answers to Q10 and Q11, to recommend using and not using groups of BoRAs for ReqElic, respectively, would probably not change with additional respondents. As a matter of fact, saturation can be confirmed also by a manual examination of the first 53 filled-in questionnaires. The answers given to Q10 and Q11 in the first 53 questionnaires covered all four categories introduced to classify the answers in the analysis of the same questions for all 92 questionnaires.

Other potential limitations to be investigated in future work include the need to work remotely due to the Covid-19 pandemic, and the emergence of new technologies. Each of them may have changed the way in which a company elicits requirements, possibly weakening the applicability of this paper's results.

6. Related work

Problems related with groups in SE and requirements elicitation have been investigated by many authors. A seminal paper focusing on the problem of people to be involved in a CBS development “The Mythical Man-Month” [6], highlighted since its first edition in 1975 the paradox that “Adding more people to a late project makes it even later.” Starting from that paper, many authors investigated groups in SE and RE. However, these studies address specific aspects of the problem, e.g., the identification of measures for team productivity [50]; or the effectiveness of requirements techniques [10], which also illustrates reasons of the shortage of empirical studies in requirements engineering.

Others addressed the problem of groups in SE for certain approaches. For example, Glass and DeMarco [17] discuss Agile approaches, suggesting that they are applicable for small projects that require 2–8 people, working in a room [7]. From this observation, one may surmise a suggestion about group sizes.

The creative nature of ReqElic is emphasized in a number of books among them those by Gause and Weinberg [14,15], by Bray [5], and by [17]. However, we found no studies providing specific, as opposed to vague, and empirically based information about the number of people to involve inReqElic.

A systematic review of the most relevant studies on groups in SE is given in [11], which highlighted that “teamwork” is one of the top research areas. A very recent review about team formation in software engineering is provided by Costa et al. [8].

An authoritative professional source for SE is the SWEBOK (Software Engineering Body of Knowledge), an international standard ISO/IEC TR 19,759:2005 developed by the IEEE Computer Science [4]. The SWEBOK – whose first part is dedicated to requirements engineering – points out that a complex CBS development project needs both group work and creativity. With regard to group RE techniques, the SWEBOK cites facilitated meetings, in which “a group of people can bring more insight into their software requirements than by working individually. They can brainstorm and refine ideas that may be difficult to bring to the

surface using interviews”. However, the SWEBOK gives no advice for group size in ReqElic.

Data about individual and group activity in software development are given by DeMarco and Lister [12]. They observe that a typical developer works alone 30% of the time, with at least one other person 50% of the time, and with two or more persons 20% of the time. In discussing the staffing of CBS development teams, DeMarco and Lister suggest specifically that to mitigate the risk of overstaffing, the initial activities of a project – analysis and design – are best carried out by smallish teams of 3–5 people. This suggestion is supported by results of a study focused on group productivity for medium-size information systems [39]. However, none of these recommendations is for the requirements elicitation activity.

There are also some empirical studies about group size for brainstorming [2], the use of focus groups [24], the impact of group size on team productivity [41].

Some authors give advice based on empirical data on recommended groups size for specific activity, noteworthy for software inspection; among them [58]. Experiments to determine how group size affects software inspection showed that even if groups of two or four inspectors were more effective than an inspector working alone, four inspectors did not find significantly more defects than two inspectors [38]. The problem with these experiments is that [23] “... without knowing the real saving gained by finding the defects, it is impossible to define the optimal team size, which would be relevant also in practice.”, a statement that recall the complexity of studies related with CBS development.

Empirical data about SE practices have been collected in the already mentioned papers by [34]; [26] and [21]. As regard the group size, [21] found that the average number of full-time staff involved in each project was 14.8. However, all these studies have a larger scope than that of RE.

7. Conclusions

The research question for this paper is as follows:

“What is a suitable size for a requirements elicitation group?”

The goal was then to find data to support requirements engineers in deciding

- 1) whether to use groups or individuals for ReqElic, the GvsI issue, and
- 2) if to use groups, what size they should be, the group size issue.

To this end, we designed and implemented two studies.

The first study was a review of most widely adopted SE and RE textbooks to see if and how they deal with the group size issue for requirements elicitation. The outcome of the SE and RE textbooks review is that an analyst will not find any real advice on whether to use groups or individuals for requirements elicitation activities, neither on the group size issue for these activities. Altogether, the final conclusion is that none of the textbooks in Table 1 and Table 2 provides real help, based on empirical data, in making decisions about the GvsI issue and about the size of a group for ReqElic.

The second part of this paper offers the results of an online survey as an attempt to remedy the current lack of empirical data. The main result of the survey is that there is no one pair of answers to the GvsI and group size issues.

In general, the explanations given by the respondents in Q9 through Q11 allow saying that

- 1) it is better to aim for small groups,
- 2) to have individual BoRAs,
- 3) but large groups are necessary in some cases.

Results of our study are about the categories of factors that are to be considered in deciding the GvsI issue in ReqElic. According to the survey, four categories can be used to help decide the GvsI issue in ReqElic.

Thus, the GvsI and group size issues must be decided in each case taking into account

- 1) the configuration of factors in the case and
- 2) the relative weights of these factors that exist for the case.

Factors in each of the four categories can help or hinder group activities in ReqElic. Among factors in the

Relations category, group communication and collaboration can help ReqElic, while conflicts and group dynamics can hinder it.

People category, skills, expertise, and other personal traits can be relevant to the effectiveness of ReqElic groups. Groups also support knowledge transfer between analysts providing some protection against the loss of a key person.

Project category, in particular those investigated in the survey, e.g., size, tightness of deadline, complexity, affect whether or not groups are used in ReqElic.

Output category, usually described in relation to quality and quantity of the requirements, are cited to justify groups because group ReqElic can potentially (1) reduce redundant, duplicated, or conflicting requirements and (2) give better requirements coverage and accuracy.

As a result, answering the GvsI issue requires dealing with the tradeoffs related to each group size. The problem can then be formulated in terms of decision making in which diverse factors have to be optimized [22]. A possible formalisation, with three configurations as proposed in Q8, and the 4 categories of factors, relative weights could be assigned as values to a 3×4 array cells [28]. Results in Table 4 could be used to assign such weights.

Future work will focus on the application of the results of the survey and of the proposed formalisation to a set of real projects for an in-depth analysis of the factors cited by the respondents. Studies of this kind are challenging, as all experimental studies in RE are. The results of the textbooks survey call the software and requirements engineering communities to find a way to facilitate analysts and researchers to share they data, also in terms of participation to online surveys like the one run for this paper. Future work could also exploit the results of the analysis given in [8,11], that review the staffing of SE groups, but focusing on ReqElic activities. Possible other work includes the analysis of other kinds of textbooks, e.g., project management textbooks, and of other kinds of sources, e.g., online posts.

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